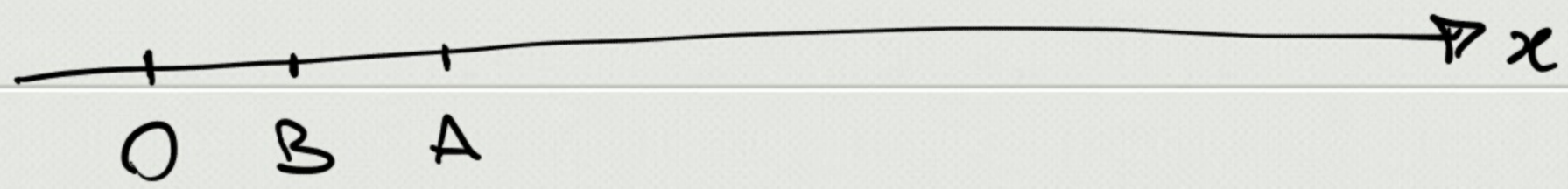


$t=0 \qquad v_{0A} = 1\ m/s \qquad a_A = 2\ m/s^2 = cost$

$v_{0B} = 3\ m/s \qquad a_B = 1\ m/s^2 = cost$

$x_{0A} = x_{0B} + 1.5 \qquad (m)$

$$\begin{cases} x_A(t) = x_{0A} + v_{0A}t + \frac{1}{2}a_A t^2 \\ x_B(t) = x_{0B} + v_{0B}t + \frac{1}{2}a_B t^2 \end{cases}$$

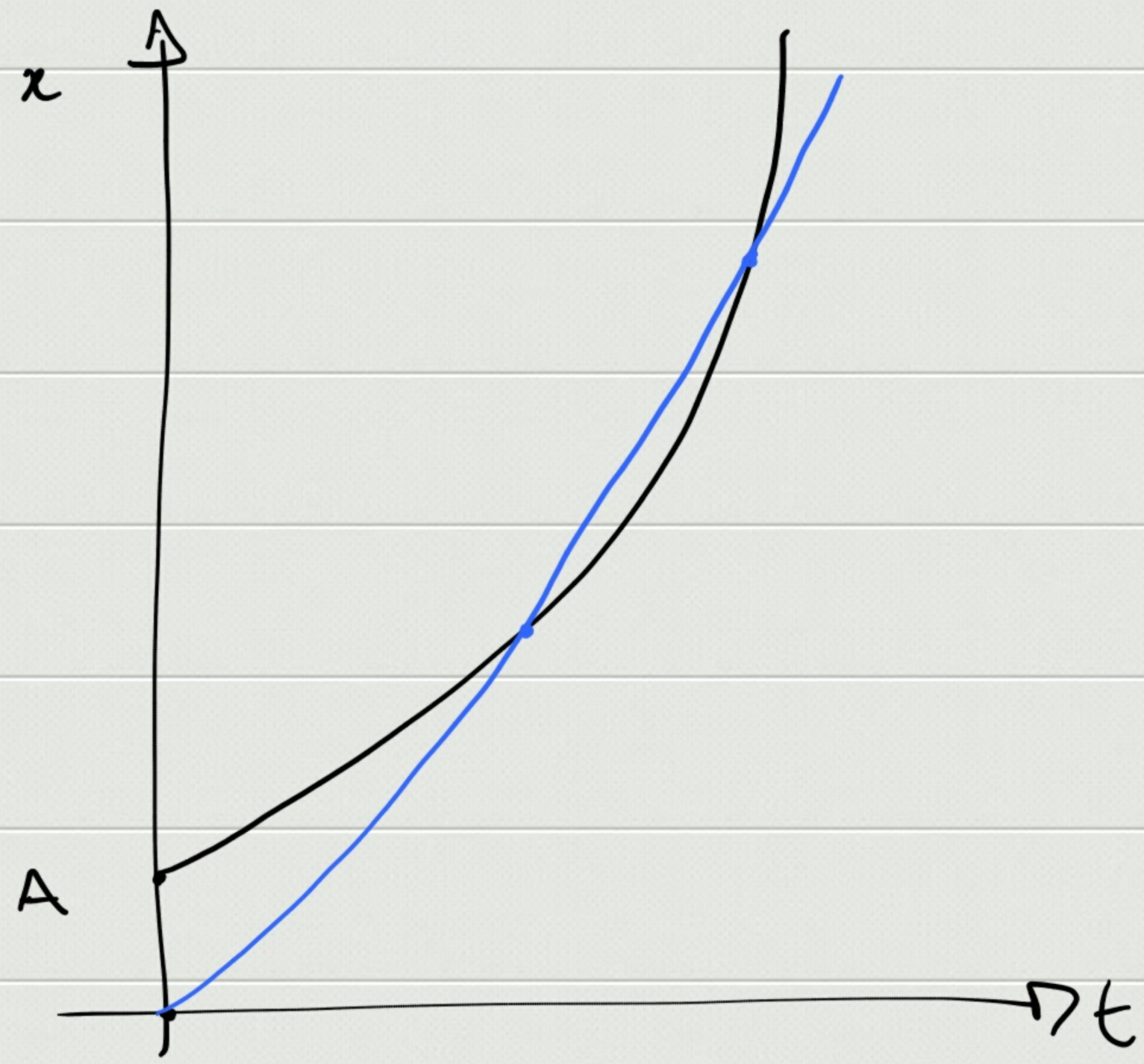
$x_A(t^*) = x_B(t^*)$

$$0 = (x_{0A} - x_{0B}) + (v_{0A} - v_{0B})t + \frac{1}{2}(a_A - a_B)t^2$$

$$t^2 + \frac{2(v_{0A} - v_{0B})}{a_A - a_B}t + \frac{2(x_{0A} - x_{0B})}{a_A - a_B} = 0$$

$$t = -\frac{v_{0A} - v_{0B}}{a_A - a_B} \pm \sqrt{\left(\frac{v_{0A} - v_{0B}}{a_A - a_B}\right)^2 - \frac{2(x_{0A} - x_{0B})}{a_A - a_B}}$$

$$t = \begin{cases} 1\ s & \Rightarrow x_1 = x_{0B} + 3.5 \quad (m) \\ 3\ s & \Rightarrow x_2 = x_{0B} + 13.5 \quad (m) \end{cases}$$


x
A
t

$v_0 = 100 \ km/h$

$a = cost \qquad (a < 0)$

$d = 100 \ m \qquad v_1 = 65 \ km/h$

$a = ?$

$$\boxed{v_f^2 = v_i^2 + 2a(x_f - x_i)}$$

- $v_1^2 = v_0^2 + 2ad \quad *$

- $v_0^2 = v_1^2 + 2ad$

- $v_1^2 = v_0^2 - 2ad \qquad (* \quad a = |a|)$

- $\begin{cases} v_1 = v_0 + at \\ d = v_0 t + \frac{1}{2} a t^2 \quad * \end{cases}$

$$a = \frac{v_1^2 - v_0^2}{2d} \simeq \underline{\underline{-2.3}} \ m/s^2$$

$$t = \frac{v_1 - v_0}{a} \Rightarrow d = v_0 \frac{\cancel{v_1} - v_0}{a} + \frac{1}{2} \cancel{a} \frac{v_1^2 + v_0^2 - \cancel{2 v_1 v_0}}{a^{\cancel{2}}} =$$

$$= \frac{v_1^2 - v_0^2}{2a}$$

$$\Rightarrow a = \frac{v_1^2 - v_0^2}{2d}$$

$d' = ?$

$v_f = v_i + a(t_f - t_i)$

- $\begin{cases} 0 = v_0 + a t_2 \\ d' = v_0 t_2^2 + \frac{1}{2} a t_2^2 \end{cases}$ *

- $v_0^2 = 2a d'$

- $0 = v_0^2 + 2a d'$ * $\Rightarrow d' = -\frac{v_0^2}{2a} = 173\,m$

- $\begin{cases} 0 = v_0 + a t_2 \\ d' = d + v_0 t_2^2 + \frac{1}{2} a t_2^2 \end{cases}$

t_2 : $t_2 = -\frac{v_0}{a} = 12{,}5\,s$